

LEDGER-8 entity structure -- ownership percentage and the parent's own Investment-in-Subsidiary account name for each

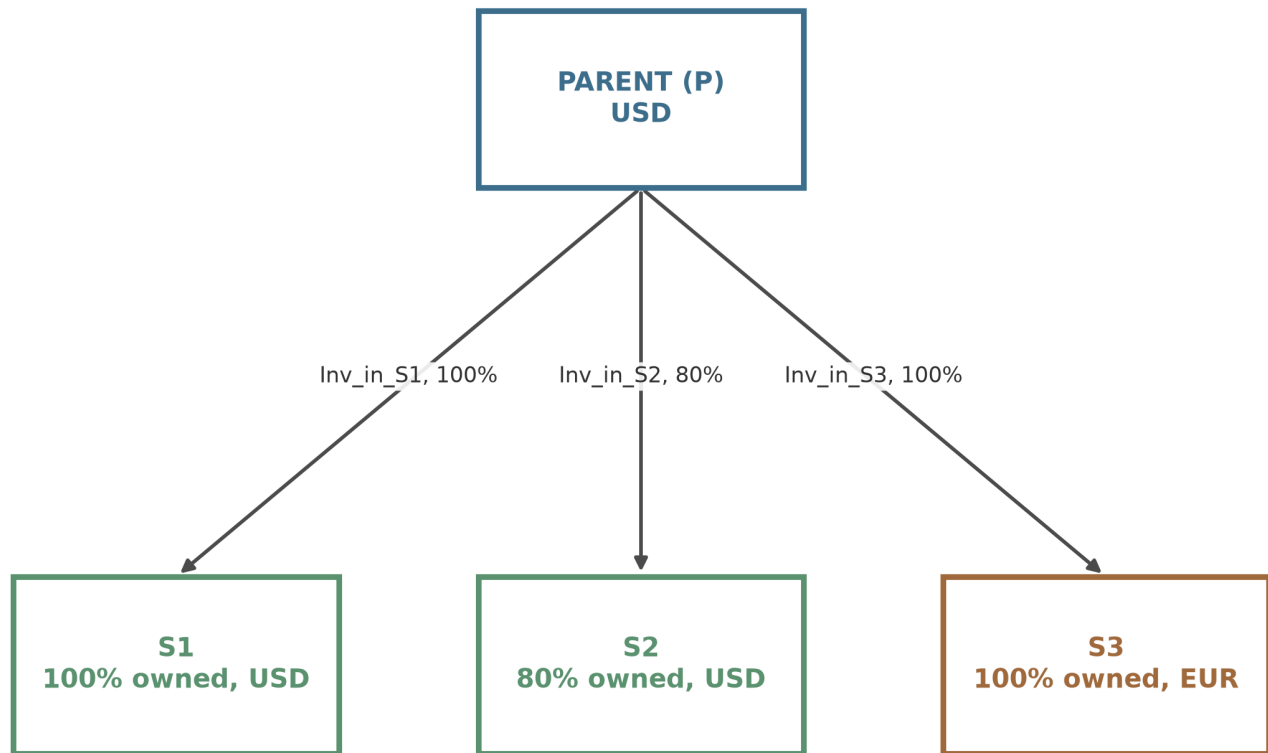


Figure 1 -- Entity structure: ownership percentage and Investment account names.

Figure 2 -- Each entity's own standalone trial balance (pre-consolidation).
RetainedEarnings is the BEGINNING-of-period balance; S3's column is in EUR.

Account	P (USD)	S1 (USD)	S2 (USD)	S3 (EUR)
ASSETS				
Cash	450,250	20,000	46,000	50,000
Accounts receivable (trade)	220,000	90,000	140,000	80,000
Intercompany receivable (trade)	45,000	--	--	--
Intercompany loan receivable	150,000	--	--	--
Inventory	150,000	70,000	110,000	60,000
Investment in S1	285,000	--	--	--
Investment in S2	316,000	--	--	--
Investment in S3	327,750	--	--	--
PP&E, net	900,000	240,000	380,000	210,000
LIABILITIES				
Accounts payable (trade)	160,000	50,000	70,000	40,000
Intercompany payable (trade)	--	45,000	--	--
Intercompany loan payable	--	--	150,000	--
EQUITY (beginning)				
Common stock	1,000,000	200,000	300,000	250,000
Retained earnings (beginning)	1,450,000	85,000	95,000	55,000
INCOME STATEMENT				
Sales -- external	1,200,000	430,000	560,000	390,000
Sales -- intercompany	100,000	--	--	--
Cost of goods sold	820,000	310,000	370,000	250,000
Operating expenses	260,000	80,000	120,000	85,000
Intercompany interest income	9,000	--	--	--
Intercompany interest expense	--	--	9,000	--
Intercompany dividend income	5,000	--	--	--

Figure 2 -- Each entity's own standalone trial balance (pre-consolidation).

Figure 3 -- S3's three EUR-to-USD translation rates (S3 reports in EUR)

Rate	EUR -> USD	Applies to
Closing rate	1.08	Assets, liabilities, common stock
Average rate	1.10	Income-statement accounts (this period's flows)
Historical rate	1.05	Beginning RetainedEarnings only

Figure 3 -- S3's three EUR-to-USD translation rates.

Figure 4 -- The four intercompany transaction facts

Inventory sale

P -> S1

Transfer price \$100,000, P's own cost \$70,000.
40% of the transferred goods remain in S1's
ending inventory at year end (unsold to any outside party).

Intercompany loan

P -> S2

Principal \$150,000, this year's interest \$9,000
(P's interest income, S2's interest expense).

Trade balance

P <-> S1

P's own intercompany trade receivable from S1: \$45,000
(unrelated to the loan above; matched by S1's own payable).

Dividend

S1 -> P

S1 paid P an intercompany dividend of \$5,000 during
the year, recorded by P as dividend income.

Figure 4 -- The four intercompany transaction facts.

LEDGER-8 -- Engineering Packet

Finance -- Corporate Accounting Consolidation

Multi-entity financial-statement consolidation and elimination-entry audit

1. ENTITY STRUCTURE (Figure 1)

One parent (P) and three subsidiaries: S1 (100% owned, domestic), S2 (80% owned, domestic), S3 (100% owned, foreign, reports in EUR). There is no time dimension and no parameter sweep -- exactly one consolidation, applied once, to one fixed set of four trial balances.

2. EACH ENTITY'S OWN TRIAL BALANCE (Figure 2)

Every entity states its own standalone trial balance (Figure 2): balance-sheet accounts plus income-statement accounts (this period's activity only). RetainedEarnings in every entity's own trial balance is always the BEGINNING-of-period balance -- this period's income-statement activity is not yet closed into it. Every entity's own assets equal its own liabilities plus its own common stock plus its own beginning retained earnings plus its own net income (an internal consistency check, independent of consolidation).

3. FOREIGN-CURRENCY TRANSLATION -- S3 ONLY (Figure 3)

S3's assets, liabilities, and common stock translate to USD at the stated closing rate. S3's income-statement accounts (this period's flows) translate at the stated average rate. S3's beginning RetainedEarnings translates at the stated historical rate -- the rate in effect when it was earned, distinct from both the closing and average rates. The Cumulative Translation Adjustment (CTA) is the resulting plug: $CTA = TotalAssets - TotalLiabilities - CommonStock - RetainedEarnings(beginning, translated) - NetIncome(translated)$, posted as its own equity line.

4. INTERCOMPANY TRANSACTION FACTS (Figure 4)

Four intercompany facts, each stated with its own amounts (Figure 4): an inventory sale from P to S1 (transfer price, P's own cost, and the fraction of the transferred goods still unsold in S1's ending inventory); an intercompany loan from P to S2 (principal and this year's interest); a separate, loan-unrelated trade receivable/payable between P and S1; and an intercompany dividend S1 paid to P during the year.

5. THE SIX REQUIRED ELIMINATION ENTRIES

Each is a genuine double-entry journal entry: converting every line to its debit-equivalent (a positive amount on a debit-normal account -- assets, expenses -- is a debit; a positive amount on a credit-normal account -- liabilities, equity, revenue -- is a credit) must sum to exactly zero. E1 eliminates the full intercompany sale (the entire transfer price) from intercompany sales revenue against cost of goods sold. E2 separately defers the unrealized profit still in the ending inventory: $(\text{fraction unsold}) \times (\text{transfer price} - \text{seller's own cost})$, via a COGS increase and matching inventory decrease -- E1 alone does not defer unrealized profit, and E2 alone does not remove the double-counted revenue; both are required. E3 eliminates the intercompany loan principal (receivable against payable) and its interest (interest income against interest expense) in full. E4 eliminates the separate, loan-unrelated trade receivable/payable in full. E5 removes the intercompany dividend from income and restores the payor's retained earnings by the same amount (an intercompany dividend must have zero net effect on both consolidated net income and consolidated retained earnings). E6: for each subsidiary, remove 100% of its own common stock and beginning retained earnings, replacing them with the parent's Investment-in-that-subsiidiary elimination at the parent's ownership percentage, and a new NCI_Equity line at the complementary (non-controlling) percentage.

6. INVESTMENT BASIS (NO GOODWILL)

Every Investment-in-Subsidiary account, as given in the parent's own trial balance, equals the parent's ownership percentage multiplied by that subsidiary's own beginning equity (common stock at the applicable translation rate, plus beginning retained earnings) -- the parent invested at each subsidiary's then-book-value, with no goodwill or purchase differential. This is what makes every Investment-in-Subsidiary account eliminate to exactly zero in E6, for every subsidiary.

7. NET INCOME AND ITS SPLIT

Consolidated net income (post-elimination) is split between the controlling interest and non-controlling interest (NCI). NCI's share is the minority ownership percentage applied to S2's own standalone net income only (S2's intercompany loan interest nets entirely against the parent's side via E3 and does not itself change S2's own reported net income). The controlling share closes into consolidated RetainedEarnings; the NCI share closes into NCI_Equity, on top of what E6 already posted for beginning NCI equity.

8. REQUIRED OUTPUT AND FEASIBILITY INVARIANT

The final consolidated balance sheet must satisfy $\text{Assets} = \text{Liabilities} + (\text{CommonStock} + \text{consolidated RetainedEarnings} + \text{CTA} + \text{NCI_Equity})$ exactly -- this is the primary feasibility invariant a submission's own work must satisfy, independent of matching any particular reference number.

9. INDEPENDENT VERIFICATION

A separately-coded auditor, sharing only immutable input constants with the primary implementation, must independently re-derive the full consolidation (or check a complete feasibility certificate) and confirm the balance-sheet balance invariant, that every one of the six elimination entries independently re-verifies to balance in debit-equivalent terms, and NCI consistency (NCI_Equity plus the controlling share of consolidated equity, summed, equals total consolidated equity net of liabilities). Two required adversarial mutations: an elimination entry that does not balance (a missing offsetting line), and a currency-translation error (translating an item at the wrong one of the three stated rates). The auditor must reject both while still accepting the true original.

10. CERTIFICATION

Build the serialization as exactly these lines, in this exact order, each line AccountName=Value with no spaces, no currency symbol, no thousands separator, no decimal point (every input and every rate is an exact decimal, so every computed amount is a plain integer -- never a rounded float), and a leading - only for a negative value:

(1) the literal header line LEDGER8-CERT-V1;

(2) one line for each of these 16 balance-sheet accounts, in exactly this order (not alphabetical), using the account name exactly as spelled here, and including every one even when its value is exactly 0:

Cash, AR_Trade, IC_Receivable, IC_LoanReceivable, Inventory, Inv_in_S1, Inv_in_S2, Inv_in_S3, PPE_Net, AP_Trade, IC_Payable, IC_LoanPayable, CommonStock, RetainedEarnings, NCI_Equity, CTA;

(3) one line for each of these 7 income-statement accounts, in exactly this order, same zero-inclusion rule: SalesExternal, SalesIC, COGS, OpEx, IC_InterestIncome, IC_InterestExpense, DividendIncomeIC;

(4) two final lines, in order: NetIncome, then NCI_NetIncome.

Example (illustrative numbers only, not this task's actual answer): if Cash were 100 and AR_Trade were 0, their lines would read exactly Cash=100 and AR_Trade=0.

Join all 26 lines (the header plus 25 account lines) with a single \n character, UTF-8 encode, hash with SHA-256; report the first 16 hex characters as the trace-integrity certificate.

Deliver: consolidation engine source; independently-coded auditor source; the final consolidated balance sheet and income statement; a findings report; certification evidence (the trace hash plus both adversarial-mutation rejection results); and an engineering memo explaining the causal chain from the intercompany facts through the six elimination entries to the final consolidated numbers. Base every reported number on your own executed implementation -- never hand-derive a balance, a total, or the hash.